Predicting the Energy Used by a House

Regression Techniques: Semester V Project

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Course: Regression Technique, Semester V

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Data Collection

- **Source:** "Appliances energy prediction" data set (UCI repository), Candanedo, Feldheim and Deramaix (2017).
- **One low-energy house.** Measured by
  - a **wireless sensor network**: temperature and humidity in different rooms;
  - **energy meters**: energy used by appliances and by lights;
  - the **nearest airport weather station**: outside weather, joined to the house data by the time stamp.
- **One row every 10 minutes**, from 11 January 2016 17:00 to 27 May 2016 18:00 (about 4.5 months).
- **Checks:** 19 735 rows, 29 columns, no missing values, no duplicated rows, no constant columns, every gap between rows is exactly 10 minutes.
- First and last day are incomplete (42 and 109 readings), so we have **136 full days** of 144 readings.

The Dataset and Why We Drop rv1, rv2

Variable	Meaning
Appliances	**Response**: energy of appliances, Wh per 10 min
lights	Energy used by the lights, Wh
T1–T9	Temperature (°C) in 9 places: kitchen, living room, laundry, office, bathroom, outside north side, ironing room, teenager's room, parents' room
RH_1–RH_9	Humidity (%) at the same 9 places
T_out, RH_out, Press_mm_hg, Windspeed, Visibility, Tdewpoint	Weather station values
rv1, rv2	Two random numbers added by the authors

	mean	sd	min	max
Appliances	97.7	102.5	10	1080
lights	3.8	7.9	0	70
T1	21.7	1.6	16.8	26.3
RH_1	40.3	4.0	27.0	63.4
T6	7.9	6.1	−6.1	28.3
T_out	7.4	5.3	−5.0	26.1

Why drop rv1 and rv2?

- They are exactly the same column, random numbers.
- Correlation with Appliances: −0.011 (p-value 0.117), so pure noise.
- We keep the other **25 covariates**.

The Original Paper and Our Target

- Candanedo et al. compared Multiple Linear Regression, Support Vector Machines, Random Forest and Gradient Boosting Machines.

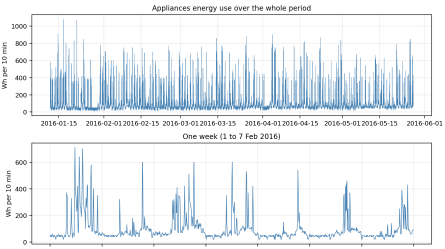
Method (original paper)	Out-of-sample R^2
Multiple Linear Regression (**worst**)	0.16
Gradient Boosting (**best**)	0.57

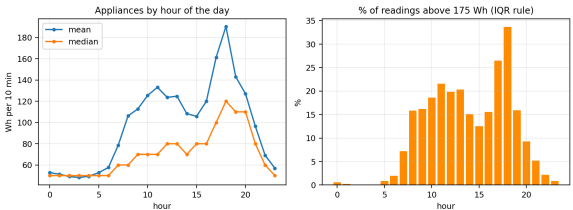
Our target

How much can we **improve** the predictive performance of the regression model, while keeping it **interpretable**?

- A model should predict reliably *and* tell us what the energy use depends on.

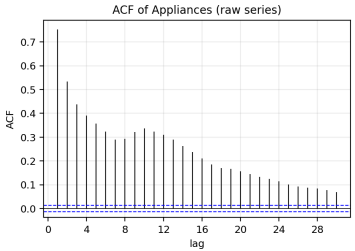
Exploratory Data Analysis: Problems with the Data

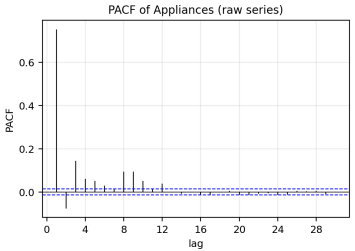




- **Strongly right skewed:** skewness 3.39 (1.17 after log); steps of 10 Wh (92 values); `lights` is 0 in 77% of readings.
- **Spikes:** base level $\approx$ 50 Wh with tall spikes; 10.8% of readings above the IQR fence (175 Wh).
- **Daily pattern:** mean $\approx$ 50 at night, 133 at 11 am, maximum 190 at 6 pm. Spikes are mostly 8 am–1 pm and 5–7 pm (33.7% above the fence at 6 pm).
- They look like real use (cooking, washing), so we **do not delete** them.

It Is a Time Series: ACF and PACF

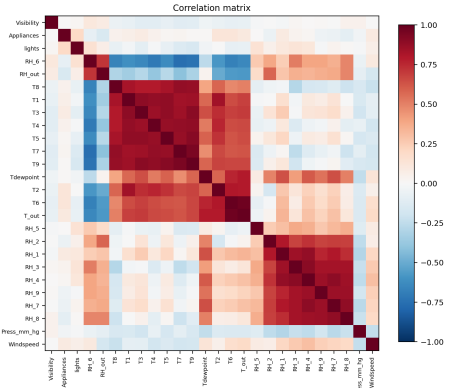


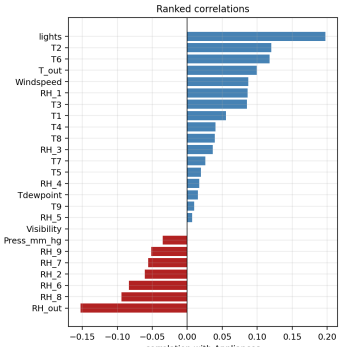


- ACF of `Appliances`: 0.75 at lag 1, 0.53 at lag 2, 0.07 at lag 30.
- Still 0.22 at lag 144, which is **one day**.
- PACF: one big spike at lag 1, small values after.

The series behaves like an **autoregressive** process, so the independence assumption of ordinary regression is in doubt from the start.

Collinearity: Correlation Matrix and Ranked Correlations





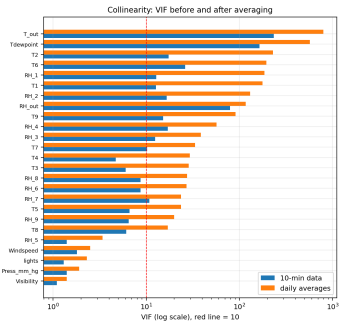
- Largest with response: `lights` 0.197, `RH_out` −0.152, `T2` 0.120, `T6` 0.118.

- Each covariate alone explains very little (none above 0.2), but covariates are strongly related to each other.
- Almost the same variable: `T6`–`T_out` (0.975), `T7`–`T9` (0.945), `T5`–`T9` (0.911), `T3`–`T9` (0.901).

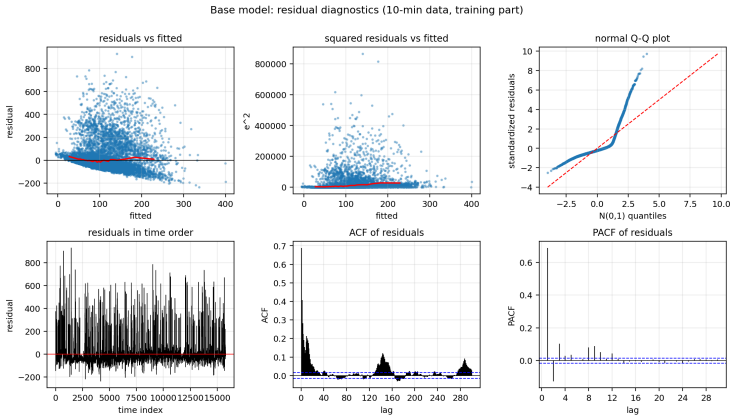
Base Model: Multiple Regression on the 10-Minute Data

$$\text{Appliances}_t = \beta_0 + \sum_{j=1}^{25} \beta_j \, x_{tj} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2) \text{ indep.}$$

- No shuffling (time series): first 80% (15 788 readings, until 30 April) for training, last 20% (3 947, until 27 May) for testing.
- Fit: $R^2 = 0.173$, adj. $R^2 = 0.172$, $F = 131.8$, $\hat{\sigma} = 95.75$.
- $\mathrm{VIF}_j = 1/(1 - R_j^2)$: largest 234 (`T_out`), 13 of 25 above 10, condition number 62.
- **Seven** significant coefficients have the *opposite sign* of their marginal correlation: `T2` ($r = +0.12$, coef -13.2); `T6` and `T_out` (corr 0.975) get $+7.6$ and -8.6: the model uses differences between almost identical variables, so single coefficients cannot be read as effects.



Base Model: Residual Diagnostics



- **Constant variance fails:** spread grows with fitted value. Breusch–Pagan $S = 4648$ (25 df), Cook–Weisberg $S' = 3980$ (1 df), p-values ≈ 0.
- **Normality fails:** Q–Q bends up (standardized residuals up to about 10); skewness 3.28, kurtosis 17.2.
- **Independence fails:** residual ACF 0.69 at lag 1 with bumps at lags 144 and 288 (one and two days). Durbin–Watson $D = 0.622$, $r_1 = 0.689$, $z = 86.5$.
- Standard errors are too small, so t and F tests are not reliable.

Base Model: Accuracy on the Test Data

	RMSE	MAE	R^2_{oos}
Base OLS (25 covariates)	86.73	52.01	0.093
Benchmark (always predict the training mean)	91.05	52.69	0

$$R^2_{\text{oos}} = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y}_{\text{train}})^2}$$

- Only **4.7% better** than the benchmark in RMSE; the MAE is almost the same.
- $R^2_{\text{oos}} = 0.093$ is about half of the in-sample 0.173.

The base model **predicts badly**, and its assumptions are violated.

Our Solutions to These Problems

The base model has dependent errors, non-constant variance, non-normal errors and strong collinearity. Two ways to go on:

Model 1: Time series model

An autoregressive model AR(p) for the 10-minute `Appliances` series.

- Deals with the dependence in time.

Model 2: Daily averages

Predict the average energy of the day from the average conditions of the same day.

- Each day is nearly one independent observation, then regression with model selection.

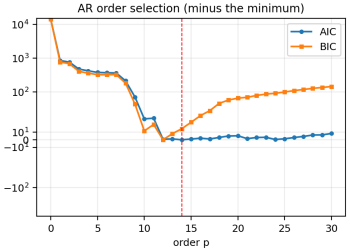
Model 1: Time Series Model (AR)

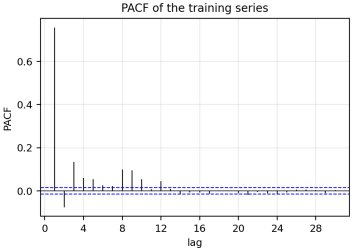
Autoregressive model of order p:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \epsilon_t$$

- Fitted on the same training part, tested on the same last 20%.
- Order chosen: $p = 14$, so **AR(14)**.
- First coefficients: $\hat{c} = 13.81$, $\hat{\phi}_1 = 0.794$, $\hat{\phi}_2 = -0.179$, $\hat{\phi}_3 = 0.084$, $\hat{\phi}_4 = 0.013$, $\hat{\phi}_5 = 0.026$.
- This agrees with the PACF: a big spike at lag 1 and small values afterwards.
- Forecasts use the true previous readings (one step ahead = 10 minutes).

How p Was Chosen



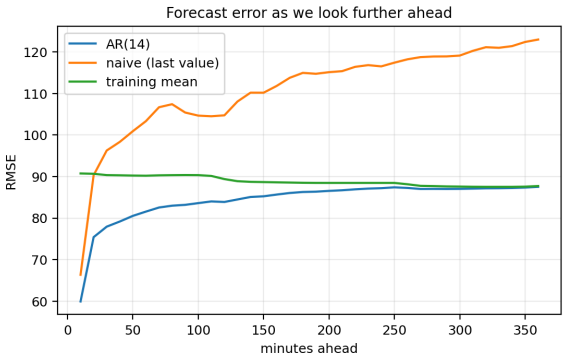


- The AIC curve falls a lot until order 10, then it is flat.
- AIC minimum at $p = 14$ (BIC would give 12).
- PACF: one big spike at lag 1, small values after.

Model 1: Evaluation

One step ahead	RMSE	MAE	R^2_{oos}
AR(14)	60.03	26.97	0.565
Naive (previous)	66.44	26.65	0.468
Benchmark (mean)	91.05	52.69	0

- Much better than the regression (RMSE 60.0 vs 86.7) and 9.6% better than the naive forecast.
- Residuals: $D = 2.05$, $r_1 = -0.026$, so lag 1 dependence is gone.
- ACF of errors still has a bump of 0.087 at lag 144: the daily cycle is not captured.



RMSE of the AR(14) forecast against how far ahead we forecast.

Problems with This Model

- It uses **only the past of** `Appliances`: no temperature, humidity or lights, so it says nothing about what the energy depends on.
- It needs the last 14 readings (**140 minutes** of data from the same house).
- It **loses its power quickly** when we look further ahead: RMSE 60 (10 min), 78 (30 min), 82 (1 hour), 87.5 (6 hours, same as predicting the mean). The naive forecast is worse than the mean after about 20 minutes.
- MAE of the naive forecast is a bit smaller (26.65 vs 26.97): the gain over repeating the last value is only in the large values.

Good for the very next reading, but it is not what we want: we also want to know **what the energy depends on**.

Model 2: Daily Averages, Idea and Model

- Predict the **average energy of the day** from the **average conditions of the same day**.
- Averaging 144 readings removes most of the dependence: each day is nearly one independent observation.
- Keep only the 136 full days (12 January to 26 May). For each day d:

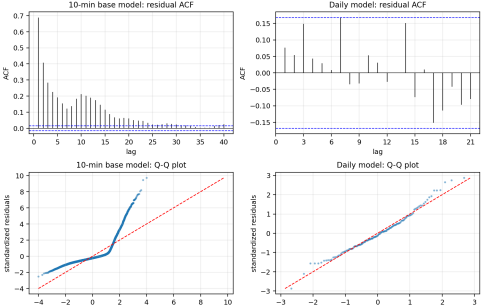
$$\bar{y}_d = \frac{1}{144}\sum_{t=1}^{144} \text{Appliances}_{d,t}, \qquad \bar{x}_{dj} = \frac{1}{144}\sum_{t=1}^{144} x_{d,t,j}$$

$$\boxed{\bar{y}_d = \beta_0 + \sum_{j=1}^{25} \beta_j \bar{x}_{dj} + \varepsilon_d, \qquad d = 1, \ldots, 136}$$

- Daily response: mean 97.4, sd 30.3, much less skewed (0.84; 0.03 on log scale).
- Covariates are averaged, never summed.

Before and After Averaging

	10-min base	Daily (25 cov.)
n	15 788	136
Adjusted R^2	0.172	0.444
Lag 1 residual autocorr. r_1	0.689	0.076
Durbin–Watson D	0.622	1.843
Durbin–Watson p-value	≈ 0	0.373
Breusch–Pagan p-value	≈ 0	0.0085
Cook–Weisberg p-value	≈ 0	0.0007
Resid. skewness / kurtosis	3.28 / 17.19	0.43 / 3.41
Skewness of response (raw / log)	3.32 / 1.13	0.84 / 0.03
Largest VIF	234	798
VIF above 10	13	20
Condition number	62.2	126.4

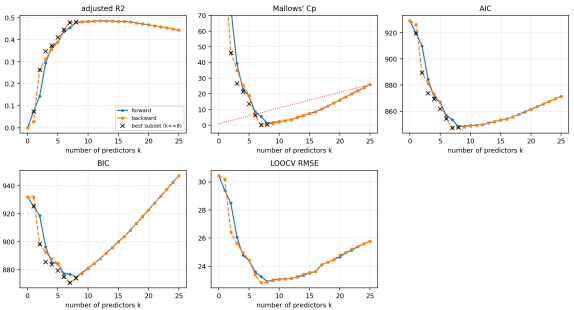


- **Better:** autocorrelation, normality. **Not solved:** constant variance (still rejected with $n = 136$). **Worse:** collinearity, so we need **model selection**.

Model Selection and Building

$2^{25} \approx 33.5$ million subsets, so we use forward and backward stepwise regression and compare sizes k with $\bar{R}^2_k$, C_p, $\mathrm{AIC} = n \log \frac{\mathrm{RSS}_k}{n} + 2(k+1)$, $\mathrm{BIC} = n \log \frac{\mathrm{RSS}_k}{n} + (k+1) \log n$ and LOOCV $= \frac{1}{n} \sum_i \left(e_i/(1-h_{ii}) \right)^2$.

Method	Criterion	Chosen model
Forward	$\bar{R}^2$	12 variables
Forward	C_p, AIC, BIC, LOOCV	8 variables (the 7 + `T_out`)
Backward	$\bar{R}^2$	same 12 variables
Backward	C_p, AIC, BIC, LOOCV	**7 variables:** `T3`, `T4`, `T7`, `T9`, `RH_1`, `RH_7`, `lights`



- BIC: backward 870.78 vs forward 874.62. All possible regressions up to $k = 8$: the best 7-subset **is** the backward model.
- F-test vs full model: $F = 0.56$ (18, 110 df), p-value 0.92, so the 18 dropped variables are not needed.

Final Model (Backward, 7 Variables)

$$\widehat{y}_d = -78.20 + 32.77\,\mathrm{T3} - 10.81\,\mathrm{T4} + 18.40\,\mathrm{T7} - 36.46\,\mathrm{T9} + 3.93\,\mathrm{RH_1} - 4.51\,\mathrm{RH_7} + 2.83\,\mathrm{lights}$$

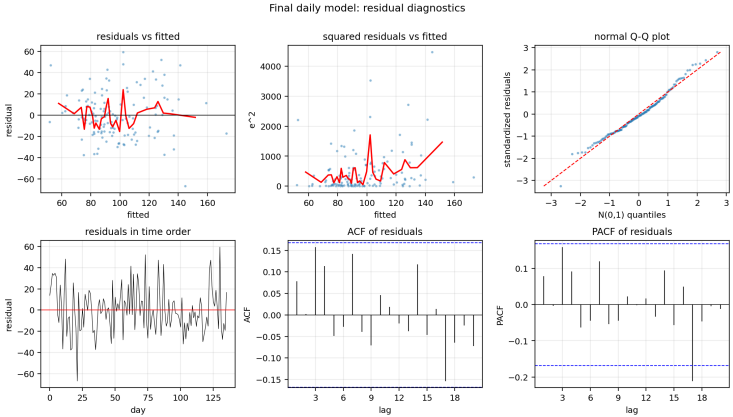
	Estimate	Std. err.	t	Std. coef.
Intercept	−78.20	32.10	−2.44	97.40
T3	32.77	3.42	9.58	62.43
T4	−10.81	3.09	−3.50	−21.02
T7	18.40	3.91	4.71	37.71
T9	−36.46	5.45	−6.69	−72.98
RH_1	3.93	1.32	2.97	14.14
RH_7	−4.51	1.01	−4.48	−20.82
lights	2.83	0.72	3.95	8.93

$R^2 = 0.505$, adj. $R^2 = 0.478$ (full model 0.444), $\hat{\sigma} = 21.9$, $F = 18.65$ on 7 and 128 df.

Interpretation

- All predictors significant (p-values slightly optimistic: selected on the same data).
- T9 (−73) and T3 (+62) matter most.
- The four temperatures have opposite signs and their standardized coefficients sum to only +6: the model uses *differences between rooms*, not "warmer = more energy".
- T9 has marginal correlation 0.010 but is the most important variable.

Final Model: Residual Diagnostics



- Residuals vs fitted: shapeless cloud; largest standardized residual 3.25.
- Q–Q close to the line (skewness 0.32, kurtosis 3.32).
- No long groups in time order; ACF/PACF inside the band at almost all lags. $D = 1.836$, p-value 0.36.
- **Weak point:** variance. Breusch–Pagan p-value 0.012, Cook–Weisberg 0.0014.
- Weighted least squares check: signs unchanged, tests no longer reject (0.41, 0.84). Mild, so we keep OLS.

Final Model: Out-of-Sample Evaluation

200 random 80/20 splits (109 training days, 27 test days); selection redone in every split on training days only.

	RMSE (sd)	MAE	R^2_{oos} (sd)	beats bench.	avg. # var.
Benchmark (training mean)	30.11 (3.54)	23.53	0	–	0
Full OLS	26.08 (3.42)	20.38	0.210 (0.318)	82.5%	25
Backward AIC	24.89 (3.35)	19.35	0.285 (0.253)	88.5%	9.3
Backward BIC	**23.93** (3.40)	**18.66**	**0.339** (0.239)	**93.0%**	6.9
Forward BIC	24.95 (3.01)	19.53	0.283 (0.233)	89.5%	6.4

Time-ordered check (train on days 1 to $s-1$, test on next 7 days; 56 test days):

	Benchmark	Full OLS	Backward AIC	Backward BIC
RMSE	30.88	31.41	28.29	27.62
R^2_{oos}	0	−0.035	0.161	0.200

- Backward BIC is best: RMSE 20.5% below benchmark, beats it in 93% of splits. The full model overfits.
- The true accuracy for new days is probably nearer to 0.2–0.3.

Combined Model: Only One Reading in the Day (Model B)

The house reports all covariates **once**, at a fixed time t_0 (3, 6 or 9 pm).

Step 1: estimate each daily average from the reading

$$\widetilde{x}_{dj} = a_j + b_j\, x_{d,j}(t_0), \qquad \hat{b}_j = \frac{\sum_d (x_{d,j}(t_0) - \overline{x(t_0)})(\bar{x}_{dj} - \bar{\bar{x}})}{\sum_d (x_{d,j}(t_0) - \overline{x(t_0)})^2}, \quad \hat{a}_j = \bar{\bar{x}}_j - \hat{b}_j \overline{x(t_0)}_j$$

A least-squares line, fitted on the training days, from the reading to the true daily mean (for each covariate j).

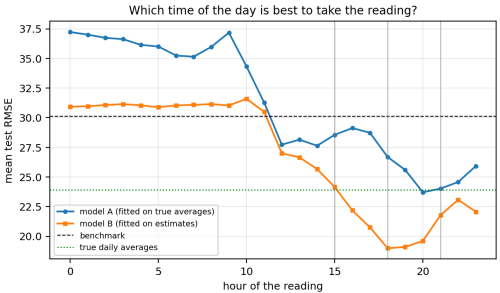
Step 2: predict the energy from the estimated averages

$$\widehat{\bar{y}}_d = \hat{\beta}_0 + \hat{\beta}_1 \widetilde{\mathrm{T3}}_d + \hat{\beta}_2 \widetilde{\mathrm{T4}}_d + \cdots + \hat{\beta}_7 \widetilde{\mathrm{lights}}_d$$

- **Model A:** the daily model of the last section, fitted on the *true* daily averages; we only feed it the estimated averages.
- **Model B:** selection and fit done on the *estimated* averages, so it learns from the same kind of input it will get.
- In every split, Step 1 lines and selection use only the training days.

Combined Model: 3 pm, 6 pm and 9 pm (Models A and B)

Read at	Model	RMSE (sd)	R^2_{oos}	beats
	Benchmark	30.11 (3.54)	0	–
	True daily avg.	23.93 (3.40)	0.339	93%
3 pm	A	28.56 (3.86)	0.033	58%
	B	24.15 (2.74)	0.334	95%
6 pm	A	26.69 (4.16)	0.155	75%
	B	**18.99** (2.79)	**0.580**	**99%**
9 pm	A	24.02 (3.64)	0.320	92%
	B	21.78 (2.74)	0.446	97%



- **Best time: 6 pm** (model B, $R^2_{oos} = 0.58$), then 9 pm (0.45), 3 pm (0.33). Lower RMSE than 3 pm in 94% and than 9 pm in 87% of splits.
- Morning readings tell nothing (RMSE flat at the benchmark from midnight to 10 am).
- B at 6 pm beats even the true-average model (0.34); a plausible reason is that the evening reflects the day's appliance use (not tested). An association, not a forecast.

Conclusion: Similar or Better than Gradient Boosting, but Interpretable

Model	Out-of-sample R^2	Interpretable?
Original paper: Multiple Linear Regression	0.16	yes
Original paper: Gradient Boosting	0.57	no
Our base regression (10-min data)	0.09	yes, but assumptions fail
Our daily 7-variable model (random / time order)	0.34 / 0.20	**yes**
Our model B, 6 pm reading	**0.58**	**yes**

- Daily averaging removes most of the dependence and the heavy tail; BIC selection gives **7 readable predictors**.
- With one reading at 6 pm we reach $R^2 = 0.58$, about the same as (slightly above) gradient boosting (0.57), with a linear model whose coefficients can be read and checked.

Indicative comparison only: we predict the daily average with our own splits; the paper predicts the 10-minute readings.

Thank You!